

ω Cen: kinematic data preparation for the dark-matter inference

oCen_dm project notes

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Abstract

This note documents how the proper-motion and line-of-sight dispersion profiles that enter the ω Cen mass inference were assembled: which catalogues, which quality selections, which estimator, and which cross-checks. The central methodological point is that both major proper-motion catalogues contain a large population of stars whose quoted uncertainties are underestimated, and that any dispersion measured from them is a statement about the error model rather than about the cluster. Once each survey’s own quality selection is respected, HST and *Gaia* EDR3 agree to 0 ± 4 per cent in their overlap, an independent Pristine-based measurement agrees with *Gaia* to 0.5 ± 1.4 per cent, and both published profiles are reproduced to a few tenths of a per cent. The final dataset is 127 binned points from five profiles spanning 0.05–57 pc.

1 Scope and principle

The inference asks whether ω Cen requires a dark component beyond its stars and remnants. That question is decided in the outskirts, where a halo would raise the dispersion above what the luminous mass supports. The outskirts are also where the data are worst. Every methodological choice below follows from a single principle, which we arrived at independently and later found stated explicitly by Häberle et al. (2025):

A star whose measurement uncertainty approaches the velocity dispersion being measured makes the inferred dispersion a function of the assumed error model rather than of the cluster.

Such stars are therefore excluded rather than modelled.

2 Data sources

Dataset	Source	Content	Role
oMEGACat VI	Häberle et al. (2025)	1.48×10^6 HST PMs to $466''$	primary inner PM
oMEGACat I / MUSE	Häberle et al. (2025)	3×10^5 LOS velocities	inner LOS
<i>Gaia</i> EDR3 members	Vasiliev & Baumgardt (2021)	2.28×10^5 stars to 0.67°	primary outer PM
<i>Gaia</i> DR2 profile	Baumgardt et al. (2019)	9 binned points	outer PM, published
Pristine periphery	Kuzma & Ishigaki (2025)	1.57×10^5 stars to 5.1°	external check
Periphery spectroscopy	Kuzma et al. (2026)	592 LOS velocities to 3.15°	external check

Table 1: Catalogues used. Only the first four enter the likelihood.

3 The dispersion estimator

All profiles measured in this project use the same estimator, applied per annulus.

Model. Each star contributes a two-component mixture. The cluster is a two-dimensional Gaussian whose principal axes follow that star’s own radial direction,

$$\mathbf{C}_i = R(\phi_i) \text{diag}(\sigma_R^2, \sigma_T^2) R(\phi_i)^\top + \mathbf{E}_i + \mathbf{D}_i, \quad (1)$$

with $\mathbf{E}_i$ the star’s full proper-motion error covariance, including the correlation coefficient, and $\mathbf{D}_i$ a rank-one term for the apparent dispersion induced by the cluster’s line-of-sight depth. The field is an *empirical* two-dimensional proper-motion density measured outside the cluster, not a Gaussian. The free parameters per annulus are σ_R , σ_T , the two mean motions, and the field fraction.

Why two dimensions. Projecting the field onto each star’s radial direction mixes its two unequal widths (3.1 and 2.0 mas yr^{−1} for ω Cen) around the annulus and manufactures shoulders that no one-dimensional model reproduces. Fitting in two dimensions removes the problem: the per-annulus χ^2 improved from 8.22 to 1.73 per bin when the field model was changed from two Gaussians to the empirical template.

Error deconvolution. Adding $\mathbf{E}_i$ to the intrinsic covariance *is* the convolution; the fit is unbinned in velocity and predicts the observed distribution directly. No subtraction is performed.

Systemic motion. The cluster’s three-dimensional velocity is projected onto each star’s own tangent basis following Vasiliev et al. (2020, §5.1), which removes the perspective contraction and the rotation of the equatorial basis across the field exactly. Subtracting a constant instead manufactures an azimuthal pattern of amplitude $|\mu_{\text{sys}}|$, which reads as a spurious dispersion of 5.3 mas yr^{−1}.

Rotation. The fitted mean radial and tangential motions absorb streaming, so σ is measured about a rotating mean. The mean-square streaming is retained so that the Jeans model, which predicts the full second moment, is compared like with like. Our fitted mean tangential motion tracks the published rotation curve of Vasiliev & Baumgardt (2021) closely: 0.226 against 0.250 mas yr^{−1} at 435″, 0.159 against 0.177 at 773″, and 0.037 against 0.036 at 1722″.

4 HST

4.1 The catalogue and its quality flag

The oMEGACat VI catalogue contains 1,482,835 sources, 1,385,010 with proper motions, reaching 466″. Its published dispersion profile stops at 300″; the catalogue does not. The flag `selection_hq_astrometry` keeps 48 per cent of the stars with proper motions and requires, simultaneously: a temporal baseline longer than 10 yr; more than 75 per cent of individual measurements surviving clipping; reduced $\chi^2 < 5$ in both components; a proper-motion error inside the lower 95 per cent of the error distribution in its own 0.5-mag bin; and reliable photometry in *both* F625W and F814W. Stars fainter than $m_{\text{F625W}} = 24$ are cut outright, because there the error limit reaches 0.3 mas yr^{−1}, comparable to half the outer dispersion.

The two-filter photometric requirement is not cosmetic: F625W and F814W span 2002 to 2022, so demanding both certifies that the astrometry is supported across the whole baseline.

4.2 Our measurement and its validation

Applying the estimator of §3 to flagged stars only reproduces the published profile across the full radial range:

r (")	N	ours (mas yr ⁻¹)	published	ratio
3.8	146	0.7708 ± 0.0382	0.8048	0.958
8.3	595	0.7944 ± 0.0225	0.7854	1.012
18.0	3224	0.7654 ± 0.0108	0.7776	0.984
39.4	15574	0.7493 ± 0.0106	0.7440	1.007
85.9	66878	0.7070 ± 0.0100	0.7079	0.999
175.8	142615	0.6268 ± 0.0089	0.6236	1.005
270.2	119576	0.5511 ± 0.0078	0.5504	1.001
310.8	16912	0.5257 ± 0.0074	0.5199	1.011

Table 2: Median ratio 1.0042, scatter 2.5 per cent, the scatter arising in the innermost bins where only a few hundred stars pass the flag.

4.3 The outer limit

The flag survives to $360''$, on 66 stars in $340\text{--}360''$ and none beyond. The reason is specific: beyond $360''$ no star in the catalogue has F625W or F814W photometry at all, so the two-filter requirement rejects them automatically. This is not a threshold that can be relaxed.

Stars failing the flag are not usable as a substitute. Running the identical estimator on each subset where both exist gives an unflagged-to-flagged dispersion ratio of 1.092, 1.088, 1.079 and 1.052 at $150\text{--}200$, $200\text{--}250$, $250\text{--}300$ and $300\text{--}340''$, a weighted mean of 1.077 ± 0.011 : their errors are underestimated by about 8 per cent. Taken raw, without a field model, they are worse still, and the apparent dispersion *rises* outwards (0.80 , 0.88 , 1.78 mas yr⁻¹ at $340\text{--}380$, $380\text{--}420$, $420\text{--}466''$), because HST proper motions are relative to the cluster so field stars sit 7.5 mas yr⁻¹ away and dominate a raw variance even at a 1–2 per cent contamination level.

5 Gaia EDR3

5.1 Why the published profile is not used

Vasiliev & Baumgardt (2021) publish a proper-motion dispersion profile on a uniform $24''$ grid from 0 to $2400''$, obtained as a cubic spline in radius with 2–5 nodes, fitted jointly with a Plummer cluster profile and a two-Gaussian field by MCMC. Five of the eight points our likelihood originally took from it lie inside $380''$, at 24, 48, 96, 168 and $336''$. Their own quality flag passes *no* star inside $200''$ and five between 200 and $300''$. The inner points are therefore an inward continuation of a stiff global model, not a measurement. Its quoted uncertainties, as small as 0.002 mas yr⁻¹, are the posterior width of a 2–5 parameter spline constrained by 10^5 stars, not the precision of any annulus.

5.2 The error model

The released catalogue carries the *raw* Gaia uncertainties. We verified this directly by joining all 228,055 members to `gaia_edr3.gaia_source`: the median ratio of catalogue to raw error is 1.0000, with 1st and 99th percentiles 0.9992 and 1.0009. The density-dependent inflation $\eta = (1 + \Sigma/\Sigma_0)^\zeta$ with $\zeta = 0.04$, $\Sigma_0 = 10$ (5-parameter) or 5 (6-parameter) is applied inside their fit, not in the file.

Deconvolving the raw errors therefore removes too little noise, and our first outer measurement ran 4–6 per cent high between 460 and $1000''$. Applying η removes the offset entirely. But η is not correct either: the authors’ own non-circular test is that a correct error model makes the deconvolved dispersion independent of magnitude, and it is not. With raw errors the faint-to-bright dispersion ratio is 1.114, 1.135 and 1.077 at $460\text{--}700$, $700\text{--}1000$ and $1000\text{--}1500''$; with η it becomes 0.954, 0.980 and 0.951. Raw errors under-correct, the density-only η over-corrects.

5.3 Our measurement

We therefore avoid depending on either. The profile uses quality-flagged stars with $\epsilon < 0.4\sigma(R)$, so that an error rescaling by η can move the deconvolved dispersion by at most $\frac{1}{2}(0.4)^2(\eta^2 - 1) \simeq 2$ per cent by construction. The quoted value is the midpoint of the raw-error and η -inflated fits and half their separation is carried as a systematic.

r (")	N	σ (mas yr ⁻¹)	stat	sys	σ_T/σ_R	f_{field}
356	52	0.4789	0.0373	0.0001	0.960	0.04
437	276	0.4497	0.0191	0.0007	0.867	0.06
533	1477	0.4350	0.0062	0.0017	0.897	0.06
662	2754	0.4016	0.0057	0.0019	0.891	0.10
830	3176	0.3614	0.0051	0.0016	0.953	0.16
1043	2895	0.3204	0.0045	0.0012	0.988	0.34
1328	2552	0.2934	0.0083	0.0008	1.083	0.59
1690	2615	0.2523	0.0089	0.0005	1.076	0.85
2148	3261	0.2314	0.0138	0.0004	1.171	0.95

Table 3: The error-model systematic is 0.1–0.4 per cent against 1.4–6 per cent statistical, so the selection achieved its purpose. The anisotropy is radial inside $\sim 1000''$ and tangential outside, which [Vasiliev & Baumgardt \(2021\)](#) report independently.

The inner edge is $300''$ because the flag passes five stars in total inside it. The two sparse inner bins are the most error-model-independent in the profile: 53 and 441 stars with errors of 0.025–0.044 mas yr⁻¹ against a dispersion near 0.47, a ratio of 5–9 per cent.

6 Cross-validation

6.1 HST against *Gaia* in their overlap

HST’s flag survives to $360''$ and *Gaia*’s begins at $300''$, so the overlap is 300 – $360''$. Three routes, differing in how HST’s rejected stars are used, agree:

Route	HST	<i>Gaia</i>	<i>Gaia</i> /HST
300–340'', both flagged	16912 at 311''	17 at 318''	1.028 ± 0.132
300–380'', both flagged	16978 at 311''	52 at 356''	0.933 ± 0.074
300–460'', HST corrected	92568 at 327''	328 at 430''	0.966 ± 0.038

The first route is the cleanest: the two samples land at $311''$ and $318''$, essentially the same place, so nothing is extrapolated and no correction is applied on either side. Within an annulus the two samples do not sit at the same radius in general, because HST’s coverage falls outwards while *Gaia*’s flag passes more stars outwards; where that matters both are moved to the annulus midpoint on the local logarithmic slope before the ratio is formed.

A star-by-star match of the two catalogues is possible but informative only about the *unflagged* population, since no quality-flagged *Gaia* star has an HST counterpart. On identical stars, unflagged *Gaia* carries 0.90, 0.79, 0.59 and 0.44 mas yr⁻¹ of scatter beyond its quoted errors at 0–100, 100–200, 200–300 and 300–460'', falling outwards exactly as crowding should.

6.2 An independent catalogue: Pristine

The Pristine periphery catalogue provides a fully independent test: different survey, different selection, different field model. Selecting on metallicity alone ($[\text{Fe}/\text{H}] < -1.2$; the field median is -0.26 against -1.49 for the cluster) and applying the same estimator with a field template built

from the same catalogue beyond 3.6° , with *no* proper-motion cut, gives a ratio to our *Gaia* profile on nine identical annuli from 9 to 57 pc of 0.995 ± 0.014 , with every bin within 5 per cent.

7 Supporting datasets

MUSE. 29 line-of-sight dispersion points from 5.7 to $290''$, taken as published. Line-of-sight and proper-motion dispersions are permitted to differ, since they probe different projections of an anisotropic velocity ellipsoid.

***Gaia* DR2.** Nine published points from 180 to $1747''$ (Baumgardt et al., 2019). Four lie inside $380''$; DR2 has its own, brighter selection and we hold no star list for it, so its usability there cannot be assessed from our side. This remains the least-controlled element of the dataset.

Periphery spectroscopy. 157 flagged members from 0.58 to 3.15° , with no proper-motion selection and therefore free of the circularity that afflicts proper-motion measurements drawn from a proper-motion-selected catalogue. They give $6.12 \pm 0.48 \text{ km s}^{-1}$ at 67 pc and 8.36 ± 1.07 at 83 pc.

8 Beyond the catalogue edge

The *Gaia* member catalogue is truncated at 0.670° (63 pc), which is the authors’ retrieval radius, not a boundary. Computed with AGAMA in the McMillan (2017) potential for $M = 3.55 \times 10^6 M_\odot$, the Jacobi radius is 90 pc at pericentre and 185 pc at the present galactocentric radius, so the data stop at $0.70 r_J(\text{peri})$.

A background-subtracted star count in the Pristine catalogue, using eight control windows in proper-motion space, detects the cluster at $78.1 \pm 10.2 \text{ stars deg}^{-2}$ at 57 pc (7.7σ), 15.9 ± 4.0 at 76 pc (4.0σ) and 2.5 ± 1.1 at 100 pc (2.2σ), falling as r^{-6} and consistent with the field beyond 130 pc in a circular average. The two spectroscopic points that carry the kinematic plateau therefore sit on a real overdensity; the third, at 178 pc from five stars, does not. Beyond 1.6° the 23 surviving candidates are not isotropic (axial Rayleigh $p = 0.016$ at position angle 148° , along-axis excess $0.34 \pm 0.16 \text{ deg}^{-2}$ against -0.02 across), which is what tidal tails look like, but the axis was chosen after inspection and the significance is 2σ .

9 The dataset that enters the likelihood

Key	Kind	N	Radial span
hst_pm_radial	PM radial	40	$1.8\text{--}311''$ (0.05–8.2 pc)
hst_pm_tangential	PM tangential	40	$1.8\text{--}311''$
muse_los_dispersion	LOS	29	$5.7\text{--}290''$
gaia_dr2_pm	PM combined	9	$180\text{--}1747''$
gaia_edr3_ours	PM combined	9	$356\text{--}2148''$

Table 4: 127 points. HST and MUSE carry everything inside 7.9 pc with 109 of them; the entire *Gaia* contribution is 18 points over 4.7–57 pc, which is the only range where a dark halo can separate itself from the stars.

10 Open decisions

1. **Extend the fitted HST range to $360''$** using our own flagged-star measurement, adding a genuine instrument overlap instead of two profiles meeting at a point.
2. **Feed *Gaia* as two datasets**, radial and tangential, as HST already is. The components are measured but the likelihood receives only their combination, discarding the anisotropy information exactly where a halo would act. This needs the per-bin covariance between the two components, which the estimator can report.
3. **Decide the status of the four *Gaia* DR2 points inside $380''$.**

References

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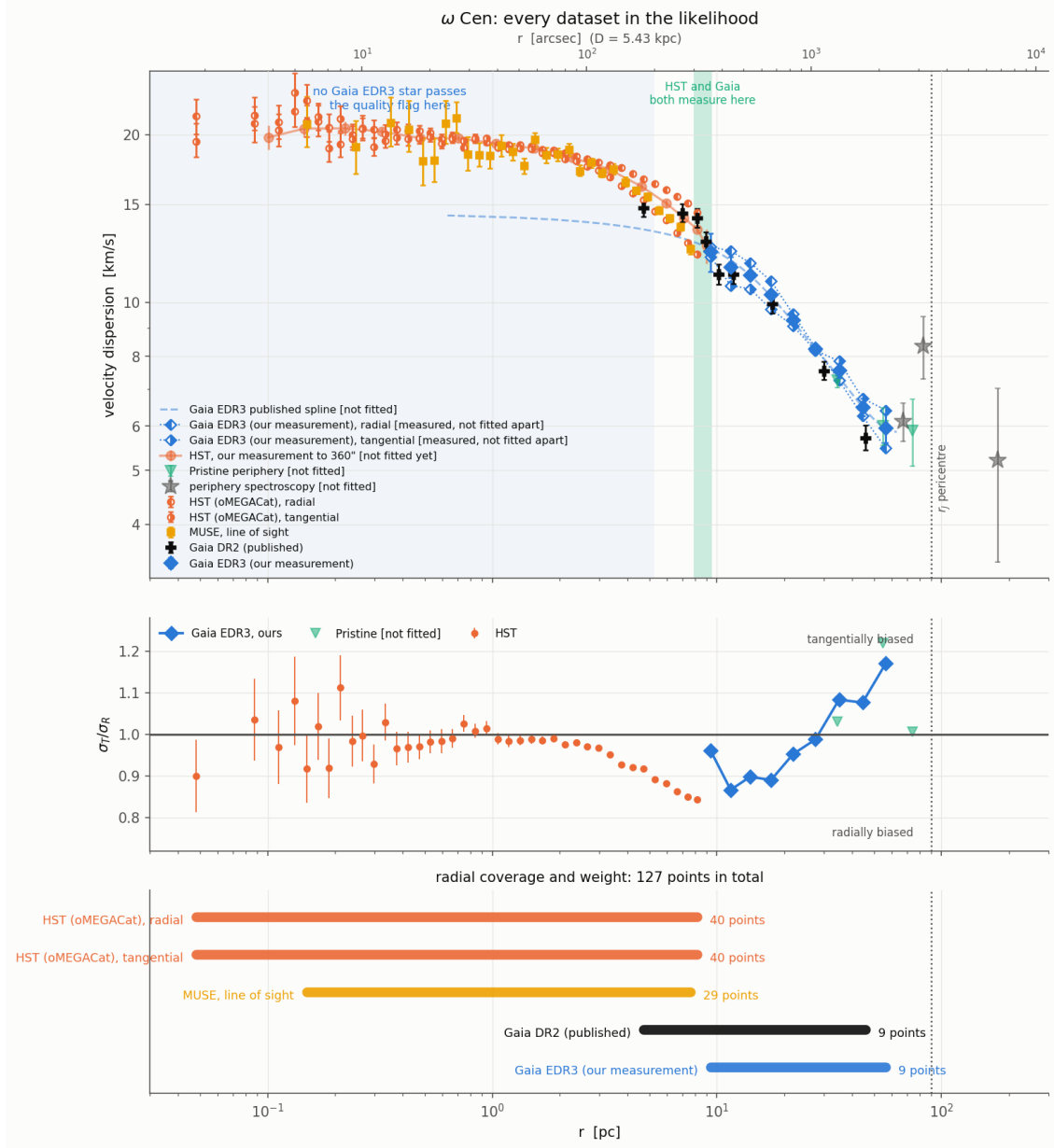


Figure 1: Every dataset in the likelihood, converted to km s^{-1} at $D = 5.43$ kpc so that proper motions and line-of-sight velocities share an axis. One symbol and colour per dataset throughout; the radial and tangential components of the same stars share that symbol and are distinguished by the marker fill. Middle: the anisotropy implied by those components. Bottom: radial coverage and point count.

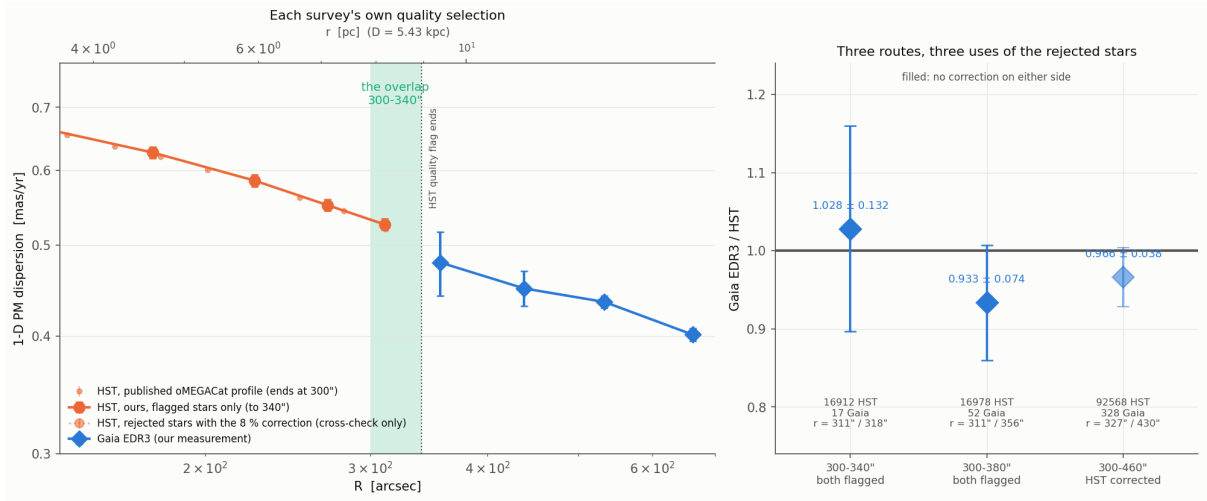


Figure 2: Left: HST measured from its own flagged stars, which reaches $360''$, against *Gaia*, which starts at $300''$. The published oMEGACat profile, which stops at $300''$, is shown faintly. Right: the comparison by three routes differing in their use of HST's rejected stars.

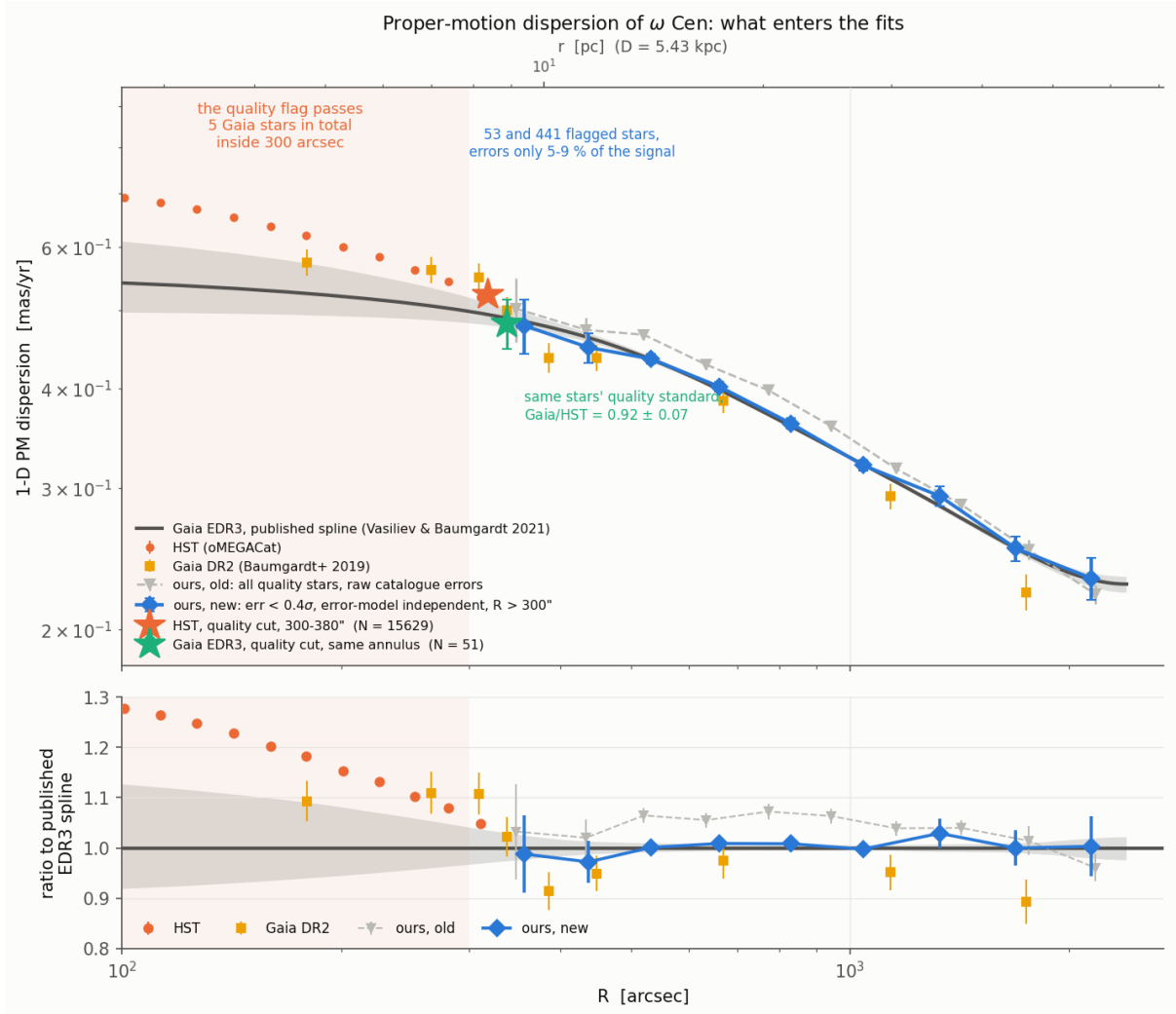


Figure 3: The rebuilt *Gaia* EDR3 measurement against the published spline and against our earlier version, which deconvolved the raw catalogue errors and ran 4-7 per cent high. The lower panel is the ratio to the published spline.

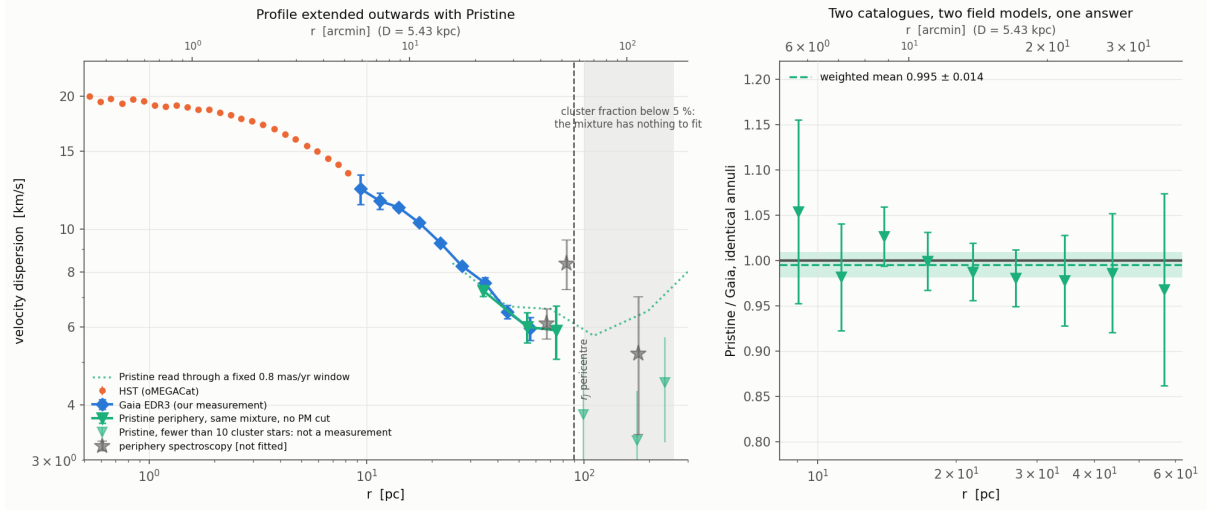


Figure 4: Left: the profile extended outwards with Pristine through the same mixture, with no proper-motion selection, against the same catalogue read through a fixed window. Right: Pristine over *Gaia* on identical annuli, 0.995 ± 0.014 .

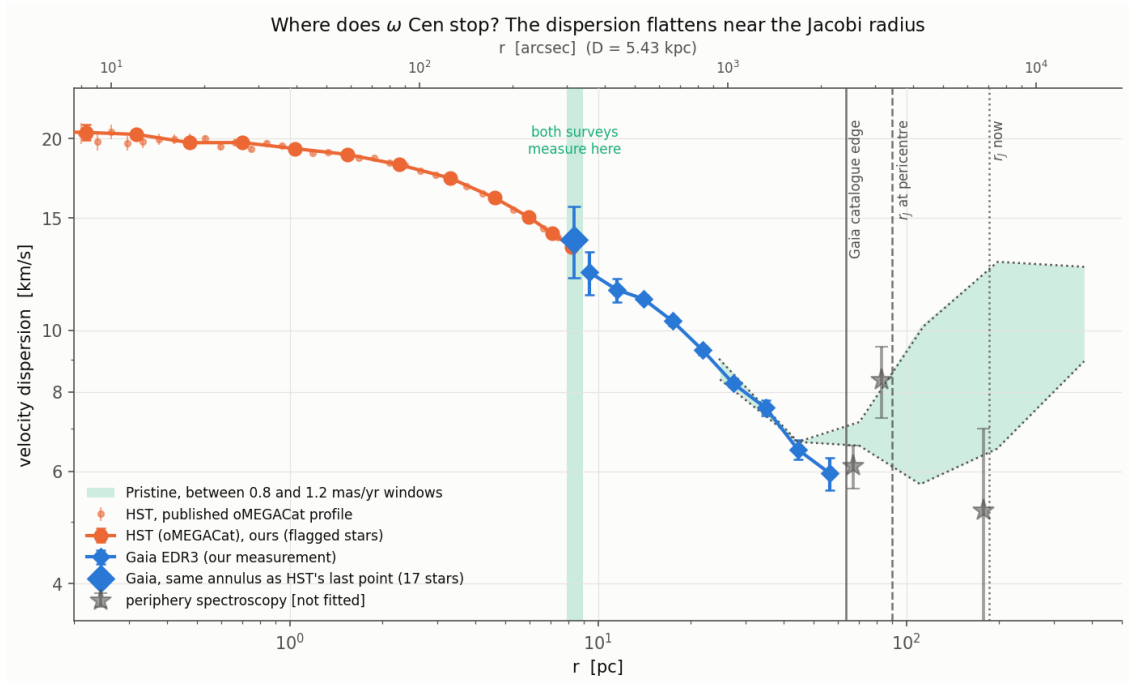


Figure 5: The profile from the centre to 400 pc. The dispersion stops falling near 50–100 pc, where the pericentric Jacobian radius lies.

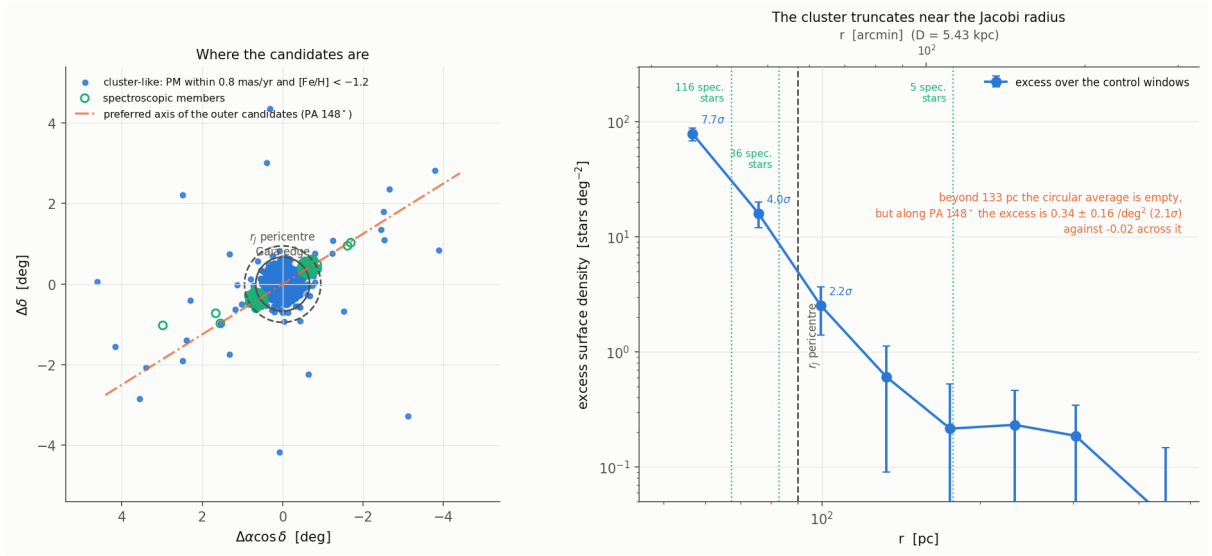


Figure 6: Left: cluster-like stars on the sky, selected on proper motion and CaHK metallicity, with the spectroscopic members overplotted. Right: their background-subtracted surface density, truncating near the Jacobi radius.